

SDH: An All-Purpose Digital Transport System

SDH is a high-speed, high-capacity, synchronous digital transport system aimed at providing a simpler, flexible and an economic telecommunications network infrastructure that facilitates technical means to transfer large quantities of data between network nodes

DR RAJIV KUMAR SINGH

The phrase 'necessity is the mother of all invention' equally applies when we talk about telecom signal transmission. Limitations of older transmission technologies, enhanced quality of service requirements, need for tightly synchronised networks and many more similar requirements are the drawing forces for the evolution of a new kind of digital transport system—the SDH (synchronous digital hierarchy).

Basically, SDH is a new kind of synchronous digital hierarchy—standardised by ITU-T (International Telecommunication Union for the telecommunication standardisation sector)—for a high-capacity telecom transmission network for transporting large amounts of voice and data traffic over a common transmission media. As the name suggests, *synchronous* implies the use of a master clock with which all network elements are tightly synchronised; *digital* implies the information in binary form and *hierarchy* implies a set of bit rates in a hierarchical order.

SDH frame structure

SDH was essentially developed to replace legacy PDH (plesiochronous digital hierarchy) systems. Its frame structure is designed in such a way that it can encapsulate earlier digital transmission standards, such as the E1, PDH or ATM (asynchronous transfer mode) networking, making SDH an all-purpose transport system. The basic unit of framing in SDH is STM-1 (synchronous transport module-level one), which consists of a matrix of nine rows and 270 columns (Fig. 1). Each cell car-

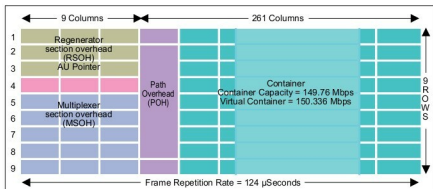


Fig. 1: Basic SDH frame structure

Different Telecom Hierarchies

European Digital Hierarchy	North American Digital Telephone System	Japanese PDH	Synchronous Digital Hierarchy (SDH)
E1 - 2.048	DS1 - 1.544	JT1 - 1.544	STM1 - 155.52
E2 - 8.448	DS2 - 6.312	JT2 - 6.312	STM4 - 622.08
E3 - 34.368	DS3 - 44.736	JT3 - 32.064	STM16 - 2488.32
E4 - 139.264	DS4 - 274.176	JT4 - 95.728	STM64 - 9953.28
(All figures in Mbps)			STM256 - 39813.12

ries one byte. Transmission is carried out row by row, starting with the byte in the upper-left corner and ending with the byte in the lower-right corner. After transmission of the last byte in the frame (the byte located in row 9, column 270) is complete, the whole sequence repeats. The frame repetition rate is 125 microseconds; therefore there are 8000 frames per second. Each byte in the payload represents a 64-kbit/s channel. So, the total number of bits that can be transported by STM-1 frames is $9 \times 270 \times 8 \times 8000$. Thus, the basic STM-1 frame bit rate equals to 155.52Mbits/s. SDH is a transport hierarchy based on multiples of 155.52Mbps in which each rate is an exact multiple of the lower rate; therefore the hierarchy is synchronous. Different SDH rates

are STM-1 = 155.52Mbps, STM-4 = 622.08Mbps, STM-16 = 2588.32Mbps, STM-64 = 9953.28Mbps and STM-256 = 39813.12Mbps.

As indicated in Fig. 1, SDH frame is made up of a container and a section overhead—section overhead being further divided into regenerator section overhead (RSOH), administrative unit pointer (AU) and multiplexer section overhead (MSOH).

The container carries customer traffic (voice, data, video, etc) and can accommodate different kinds of signal data like Ethernet, PDH signals and ATM signals. The container, together with path overhead (POH), makes a virtual container (VC). POH is used for end-to-end signaling and error measurement.